**Math 1271 - Worksheet 9** *10/8/2026*

## 1.8 - Linearization

1. Let $f(x) = \sqrt{x}$.

   (a) Find the linearization of $f(x)$ at $a = 1$.

   (b) In class, we used the linearization of $f(x)$ at $a = 4$ to approximate the value of $f(3) = \sqrt{3}$. Suppose that we instead use the linearization at $a = 1$ to do this. *Without* computing the resulting approximation, do you expect this approximation to be better or worse than the one given in class? (Hint: Think geometrically.)

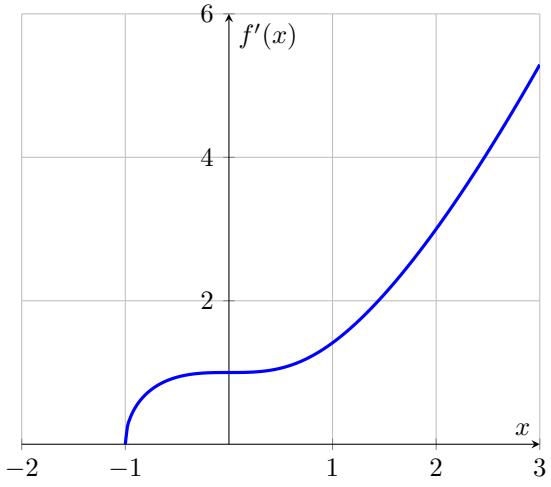
   (c) Use the linearization of $f(x)$ at $a = 1$ to approximate the value of $f(3) = \sqrt{3}$.

2. Suppose that $f$ is a differentiable function and that $f(2) = -3$. You may also assume that $f'(x) = \sqrt{x^3 + 1}$.

   (a) What is the linearization of $f(x)$ at $a = 2$?

   (b) Use your answer to part (a) to estimate $f(2.5)$.

   (c) A graph of $f'(x)$, the derivative of $f$, is shown below. Based on this graph, is your answer to part (b) an underestimate or an overestimate of the true value of $f(2.5)$?



3. Suppose it is known that for a given differentiable function $y = g(x)$, its linearization at $a = -1$ is given by $L(x) = -2 + 3(x + 1)$.

   (a) Find $L(-1)$ and $L'(-1)$. How do these compare to $g(-1)$ and $g'(-1)$?

   (b) Do you expect the value of $g(-1.03)$ to be greater than or less than the value of $g(-1)$? Why?

   (c) Use $L(x)$ to estimate the value of $g(-1.03)$.

(4) Use relevant derivative rules to differentiate each function below.

(a) $h(z) = \sqrt{z} + \dfrac{1}{z^4} + 5^z$ (Please use Leibniz notation.)

(b) $r(x) = \dfrac{3^x}{x^4 + 1}$

(c) $p(t) = (t^3 - t^2 + t - 1)(t^2 + 3t + 1)$

(d) $g(y) = \dfrac{y^2 e^y}{2y + 1}$

(5) Suppose that $f$ and $g$ are differentiable functions such that

$$f(4) = 2, \qquad f'(4) = 6, \qquad g(4) = 5, \qquad g'(4) = -3$$

Using this information, compute the derivatives of each of the following functions:

(a) $h(x) = f(x)g(x)$. What is $h'(4)$?

(b) $S(t) = \dfrac{f(t)}{g(t)}$. What is $S'(4)$?

(c) $q(z) = z^2 f(z) + z$. What is $q'(4)$?

(d) $k(x) = \dfrac{x}{f(x)}$. What is $k'(4)$?

(6) Suppose that a bakery sells cakes. Each cake is priced at $p = 20$ dollars. At this price, the company sells 500 cakes per month and demand is falling at a rate of 30 cakes (sold per month) per dollar.

The bakery is looking over its finances. In particular, it is looking at its *monthly revenue*:

$$\text{Monthly Revenue} = \Big(\text{Price per cake}\Big) \times \Big(\text{Number of cakes sold per month}\Big)$$

The bakery would like to increase its monthly revenue. To do this, the bakery is considering one of the following options:

- increase the price of a cake to $p = 21$ dollars;
- keep the price of a cake at $p = 20$ dollars;
- decrease the price of a cake to $p = 19$ dollars.

In this problem, we'll use calculus to help us determine what the bakery should do. Let $f(p)$ be the number of cakes sold (per month) when the price of a cake is set to $p$ dollars.

(a) How can the statement "when each cake is priced at $p = 20$ dollars, the bakery sells 500 cakes per month" be rephrased in terms of the function $f(p)$ and/or its derivative, $f'(p)$?

(b) How can the statement "when each cake is priced at $p = 20$ dollars, demand is falling at a rate of 30 cakes per dollar" be rephrased in terms of the function $f(p)$ and/or its derivative, $f'(p)$?

(c) Let $R(p)$ be the monthly revenue (in dollars) that the bakery earns each month from the sale of cakes. What is a formula for $R(p)$ in terms of $p$ and $f(p)$?

(d) Use the Product Rule to find a formula for $R'(p)$ in terms of $p$, $f(p)$, and $f'(p)$.

(e) At what rate is the monthly revenue changing when the price of a cake is set to $p = 20$?

(f) Based on your answer to the previous part, what should the company do? Should it raise the price of a cake? Keep it the same? Or decrease the price of a cake? What would lead to higher monthly revenue? Why?